Birla Institute of Technology & Science, Pilani
Pilani Campus
AUGS/AGSR Division

GPIC 2026 | Gujarat Police Innovation Challenge

Module 01 — PRAHARI

Lectures Stakeholder briefing

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Outcomes

- Tell the school-notebook story: cameras already exist, the watchlist is unused
- Name the hybrid (Models 1 + 2 + thin 3) and say why Model 4 is Phase-2
- Trace one frame through capture, analyse(), matcher, alert, CSV
- Separate MEASURED laptop facts from DESIGN TARGET 80k / 5–6 Cr numbers
- State the FRS law: enrolled gallery on Own cameras, never Paldi Circle

Who this briefing is for

- Home Department evaluators, departmental operators, the student team
- Voice: a lecturer naming parts a ten-year-old can follow
- Every number is MEASURED, DESIGN TARGET, or unused (CONJECTURED)
- Page: `https://sentinel.gujarat.gov.in/problems`
- Category 1 student. Lock 07 September 2026, 12:00 IST

The school-notebook analogy

- 26 schools. Each bought a different cupboard for tapes
- Some cupboards keep 7 days. Some keep 15
- Head office has a notebook of stolen bicycles and missing children
- Nobody in the camera room opens that notebook
- That is Gujarat CCTV: 26 departments, analog and IP, 1,000 km
- Watchlists exist (VAHAN, SARTHI, eGujCop, AFIS, NAFIS) and sit unused

What the State already has

- Independent CCTV estates. No statewide camera census
- Cloud NVR in some estates, local storage in others
- Sites: border districts, Valsad, Dahod, Somnath, Jamnagar, Dwarka
- The official goal: use existing infrastructure first
- A new cupboard for every tape is the wrong first move

Four official challenges

- Heterogeneous infrastructure: vendors, VMS, AMC, formats, protocols
- Geographical dispersion of about 1,000 km
- Unified analytics and event handling across onboarded cameras
- New cameras and departments without a redesign

Core goal on the official page: use existing infrastructure to the maximum practical extent.

Reference models (official)

Model	Official name	School picture
1	Registry and GIS	Roll call. No live picture required
2	Unified viewing	One window, many classrooms
3	VMS federation	Post office between cupboards
4	Central VMS	One giant cupboard for the city
H	Hybrid	Combine two or more

- Model 1 is mandatory foundation for 2, 3, and 4

Options we rejected

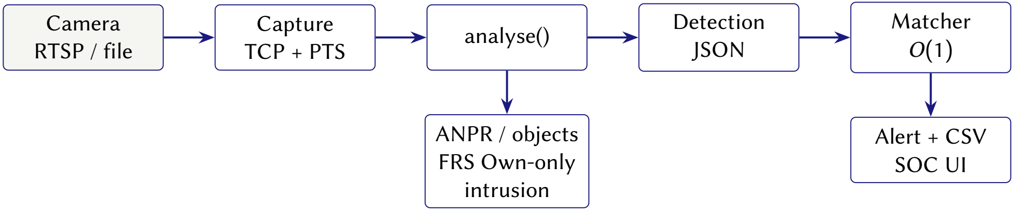
- Model 1 only: a roll call does not catch a stolen car
- Model 2 only: viewing without a watchlist fails the test case
- Full Model 3 SDK zoo: seven days cannot wrap every vendor
- Model 4 statewide: we cannot host 80,000 recordings (DESIGN TARGET)
- Wild FRS on Paldi Circle: unknown members of the public
- A ministry biometric or vehicle API: we have none. Claiming one is false

Chosen model

Layer	Official model	PoC
Census, GIS, health, gaps	1	now
Unified viewing + analytics	2	now
Event bus, watchlist, track	thin 3	now
Central VMS recording	4	Phase-2, selected cameras

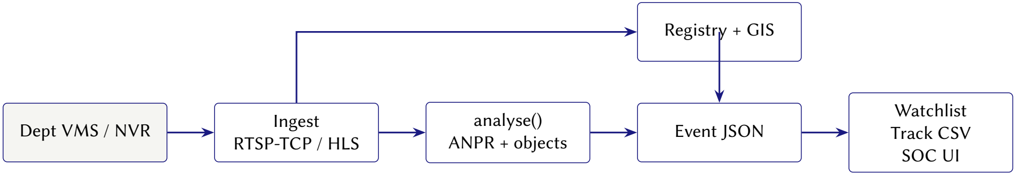
- A new vendor is one adapter on RTSP / HLS / WHEP / ONVIF
- Detection JSON does not change if the bus later becomes Kafka

How a picture becomes a red card



- Capture stamps PTS. `analyse()` samples about 1 fps
- Matcher is $O(1)$. Stolen plate raises CRITICAL. CSV keeps timestamps

Logical architecture



- PoC bus is an in-process WebSocket. The JSON is the contract
- Analytics samples 1 frame per second, not a 25 fps GOP

Official test case

- Onboard about 50 heterogeneous cameras from the Resources page
- Integrate live feeds. Enable monitoring and video analytics
- Identify a designated vehicle registration number across the grid
- Reconstruct route, timestamps, locations. Plot on GIS
- Match detections to a representative watchlist. Raise automated alerts
- Output: working platform, not a mock. CSV / report with timestamps

Live Sentinel contract (03 Sep 2026)

- Portal: `cctv.corp8.cloud` after access-password login
- Catalogue: `GET /cameras.json` returns 30 rows `{id, name}`
- `/api/ingest` is HTTP 404 on this host
- HLS: `https://cctv.corp8.cloud/<id>/index.m3u8` (cookie + browser UA)
- RTSP: `rtsp://103.250.160.189:8554/stream/<id>` (TCP)
- One second of video takes one second to arrive. There is no file download

Integrator laws in the capture client

- Set `OPENCV_FFMPEG_CAPTURE_OPTIONS=rtsp_transport;tcp` before `import cv2`
- Event time is PTS (`CAP_PROP_POS_MSEC`), never arrival time, never declared FPS
- Reconnect backoff starts at 2 s and caps at 30 s
- Decoder warnings at join are logged, not fatal
- PTS rewind or gap > 5 s is a scene cut (feeds loop)
- Consume only. Do not wget /stream/<id>
- Tests: `tests/test_integrator_laws.py`

Why those laws (ten-year-old version)

- UDP across a firewall arrives as a broken picture that looks like a bad model
- After join, extra frames may arrive fast. Wall-clock time invents flying cars
- Feeds restart. Reconnecting in a tight loop is shouting at a locked door
- The tape loops. At the splice, forget the old object ids
- A downloaded `/stream` file is a fake dataset

MEASURED: government feed (03 Sep 2026)

- Synced 30 catalogue cameras into the registry (`cam01` ...`cam30`)
- RTSP-TCP probe live: `cam04` Paldi Circle, `cam01` Chimanbhai Bridge
- First live frame: `03_Data/recordings/first_live_frame.png`
- Size 1920×1080, PTS 1080 ms. UDP SETUP 461 then TCP succeeds
- Manifest has no `live` flag and no lat/lon. Pins stay off the map
- Open tiles from the Cameras table, not from a 0,0 marker

MEASURED: designated vehicle

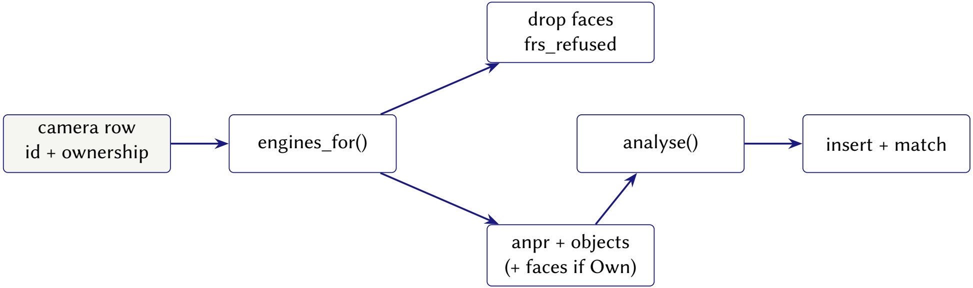
Time	Camera	Location
06:12	CAM-VAL-001	Valsad NH-48 Toll
07:48	CAM-SUR-001	Surat Adajan Circle
09:05	CAM-AHD-004	Narol–Naroda Hwy
09:22	CAM-AHD-001	SG Highway Junction
09:51	CAM-GNR-003	Koba Circle
10:04	CAM-GNR-001	CH-0 Sector 21 GNR
11:40 IST	cam04	Paldi Circle (live confirm)

- Plate `GJ01AB1234`, watchlist `WL-001` / STOLEN
- Live confirm appends. It does not drop the six seed points

`analyse()`: one door

- Default engines: `anpr,objects`
- Own cameras may add `faces`
- `cam` plus digits, and any `Gov` camera, never run FRS
- Godown `CAM-FCS-001` also runs person-in-ROI intrusion
- Plate fields stay frozen. New fields are additive
- Ingest writes crops, inserts SQLite, then calls the matcher

`engines_for()`: who may run faces



- Log line `frs_refused` is the control, not a post-filter
- FaceAnalyzer is not constructed on Paldi Circle
- `POST /api/ingest/frame` stays ANPR-only

ANPR: options

- Train a detector from scratch: no labelled Gujarat set in seven days
- Require GPU YOLO now: GPU count MEASURED 0 on this laptop
- Morphology + Tesseract behind `recognize()`: CPU, swappable
- Operator confirm: a readable plate cannot fail a live demo
- Confirm writes `source=operator_confirm`. Never labelled ANPR

ANPR: mechanism

- Sampled frame → plate box → Tesseract → Indian normaliser
- Regex after stripping space and hyphen: `^[A-Z]{2}[0-9]{1,2}[A-Z]{1,3}[0-9]{3,4}$`
- Interface: `recognize(frame) -> {plate, plate_raw, confidence, crop}`
- Missing Tesseract binary: empty plate, no crash
- MEASURED 03 Sep: confirm on `cam04`, confidence 1.0
- Optional: `ANPR_ENGINE=yolo` vehicle-crop then Tesseract, then fallback

Objects: options

- YOLOv8n as the only path: extra dependency, tests would skip
- OpenCV DNN with large weights: allowed later, not required for pytest
- Skin-tone blob fallback: a rectangle $\geq$ 8% of the frame is a person
- Skipping objects: eval area 05 names person and vehicle detection
- Classes kept: person, car, motorcycle, bus, truck, bicycle

Objects: mechanism

- Default `OBJECT_ENGINE=opencv`: blob, then optional DNN
- Optional `OBJECT_ENGINE=yolo`: COCO subset including person (class 0)
- Missing ultralytics or weights: blob fallback, no crash
- Each box gets per-camera `track_id` (IoU default)
- Scene cut calls `objects.reset`
- Evaluator artefact: `GET /api/objects/report.csv`

Intrusion on a godown

- Food and Civil Supplies watches a warehouse, not a highway
- ROI lives in `cameras.extra_json`
- Person box overlapping the ROI by $\geq 30\%$ of its area fires
- Event: `entity_type=intrusion`, CRITICAL
- Same camera within 120 s increments a counter
- No fourth neural net. Intrusion wraps object boxes

FRS: options we refused

- No FRS: HLD text and eval 05 name recognition
- Wild FRS on Paldi Circle: unknown members of the public
- Live AFIS / NAFIS: we have no ministry API. Claiming one is false
- Enrolled gallery on Own cameras: consented adults or synthetic fixtures
- Human confirm-face exists, same safety net as plate confirm

FRS law (non-negotiable)

- Own-feed may show a gallery match on `CAM-OWN-001`
- Government CCTV of unknown people never runs FRS
- Tests post a face-like still to `cam04` and expect zero person events
- Gallery JSON has no embeddings
- Spoken line: representative person watchlist, not a NAFIS join

FRS: default histogram mechanism

- Crops live under `data/faces/{gallery_id}/`
- Match: colour histogram of a 64×64 crop, correlation score
- Threshold: `FACE_MATCH_MIN_CONFIDENCE` (default 0.55)
- `WL-004` fires on `face_id`, not on an empty plate
- Synthetic fixtures are ellipses with unique noise
- Never a minor. Never a crop from `cctv.corp8.cloud`

Optional vision engines

- FaceNet when `FACE_ENGINE=facenet`, Own cameras only
- YOLO when `OBJECT_ENGINE` or `ANPR_ENGINE=yolo` (person class included)
- ByteTrack when `TRACK_ENGINE=bytetrack`; scene cut still resets
- Next-camera predict: frequency + distance, not Kalman
- Never Paldi FRS. No CodeFormer in this PoC
- Default remains histogram + blob + Tesseract + IoU
- Workshop: ArAv-1/PRAHARI-3.0. Submission repo remains amitduabits/PRAHARI

FaceNet door (optional)

- Lazy `FaceAnalyzer`: MTCNN boxes + 512-d InceptionResnetV1
- Cosine against `.npy` files next to gallery JPEGs
- Torch is imported inside `__init__`, never at package import
- Missing torch: histogram fallback, no crash (T-V01)
- Constructed only after `engines_for` allows faces (T-V04)
- `requirements-vision.txt` is optional. Pytest does not need it

YOLO and ByteTrack doors (optional)

- Object YOLO includes person. Vehicle YOLO for ANPR does not
- Weights: `app/models_data/yolov8n.pt`, gitignored
- Fetch script exists. Pytest does not download
- ByteTrack writes object `track_id`. IoU remains default
- `GET /api/track/{plate}` stays the designated-vehicle test
- GPU count MEASURED 0 on this laptop

Crop honesty

- Always write `crop_uri_original`
- Matcher sees original pixels only
- Optional enhanced copy: `enhancement_method=cubic_upscale` or `none`
- Face crop with `is_ai_reconstructed=1` $\rightarrow$ `pending_review`
- Reconstructed faces do not auto-fire CRITICAL
- Zero branded reconstructor names in `app/` (T-V11)

Watchlist and alerts

- Representative sources. No live ministry API in the PoC
- $O(1)$ on plate and on `face_id` / `gallery_id`
- CRITICAL stolen/wanted/intrusion. HIGH blacklist or person associate
- Same entity + same camera within 120 s increments a counter
- WebSocket `/ws/alerts` plus 5 s poll fallback
- `WL-004` fires without a plate. Person cards show a name, not a blank
- MEASURED: confirm on `cam04` raised CRITICAL for `GJ01AB1234`

Person enroll and pending review

- Watchlist form: Enroll Missing/Wanted Person
- Posts `/api/faces/enroll`. Viewer still 403 (T-V10)
- Seven tabs. No eighth top-level tab
- Alerts renderer shows the word `pending_review`
- Ack still works
- Do not enroll from a Sentinel still

Vehicle track is plate GIS, not Re-ID

- `GET /api/track/GJ01AB1234`: six seed cameras in order
- Live confirm appends. Seed ids stay
- Teleport flag if implied speed > 180 km/h (information, not a fail)
- CSV is the evaluator artefact
- ByteTrack does not replace this API

Next-camera predict (bonus)

- `GET /api/predict/{plate}`, auth required
- Rank historical transitions out of the last camera
- If none: GIS neighbours by haversine
- Method field: `frequency+distance`
- Not Kalman. Not Re-ID
- Track tab button: Next cameras

Keyword query (optional)

- `POST /api/query` extracts plate, camera id, object class
- Response includes `engine: keyword_rules`
- Regular expressions, not a language model
- T-V09 forbids `nlp` and `llm` in the body
- Do not put "natural language AI" on a slide

Registry, GIS, gaps

- CSV, form, REST, catalogue sync
- Dahod GSRTC is seeded offline on purpose
- Short retention: Food and Civil Supplies 7-day cameras
- Private-permitted mall requires `consent=true`
- `home.viewer` never sees that mall
- Live catalogue has no coordinates. Open `cam04` from the table

Seven tabs a judge will click

- Operations: map, health pins, wall of at most four tiles
- Cameras: table Open tile (catalogue has no GIS)
- Vehicle Track: reconstruct, CSV, Next cameras
- Alerts: CRITICAL queue, `pending_review`, Ack
- Watchlist: representative rows, person enroll
- Onboard: ANPR this still, Analyse this still, confirm
- Analytics & Gaps: object CSV, short retention, Dahod

Evaluation map A (official areas 1–4)

Official area	PRAHARI now
1 Test case on gov feed	30 cameras onboarded. RTSP-TCP frame. Confirm + CSV on `cam04`
2 Solution presentation	This deck + notes. Hybrid justified
3 Architecture	HLD. Open adapters. Model 4 not faked
4 Working platform	FastAPI on :8080. Own-feed file + live catalogue

Evaluation map B (official areas 5–7 and bonus)

Official area	PRAHARI now
5 Analytics output	ANPR/confirm + objects CSV + godown intrusion + own-feed gallery FRS
6 Scale / PoC ready	1 fps regional DESIGN TARGET in HLD §5 / §10
7 Completeness	GitHub live. YouTube and Drive still human
Bonus (on camera)	Hybrid, track, objects, intrusion, lawful FRS, RBAC, gaps, WS

Test families (04 Sep 2026)

- Default engines: 88 passed, 4 skipped. `audit_gate.py` PASS
- Integrator: TCP, no `CAP_PROP_FPS`, scene-cut resets track ids
- ANPR: normaliser, junk drop, confirm $\neq$ `source=anpr`
- Objects: person blob inserts. YOLO without weights does not crash
- FRS: same gallery matches. `cam04` emits zero person events
- Matcher: stolen, blacklist, `WL-004` without a plate, 120 s dedupe
- Security: viewer 403 on enroll. Auditor 403 on confirm-face. No RTSP in JSON

Vision tests T-V01 to T-V11

- T-V01 / T-V02: missing torch or ultralytics → histogram / blob, no crash
- T-V03 / T-V05 / T-V06: skip unless extras and weights exist
- T-V04: FaceAnalyzer not constructed on `cam04`
- T-V07: original crop uri. Reconstructed → `pending_review`
- T-V08: predict returns a list. T-V09: `keyword_rules`
- T-V10: Enroll in index. Viewer 403
- T-V11: zero forbidden model names in `app/`

Experiments: MEASURED versus SKIPPED

- E-O1 person blob, E-F1 gallery, E-I1 intrusion, E-W1 track ≥ 6: MEASURED
- E-S1 fifth capture rejected. E-S6 GPU 0. MEASURED
- E-A1 Tesseract fixture: SKIPPED if binary absent (confirm covers demo)
- E-G1 live catalogue: SKIPPED if `SENTINEL_HOST` empty
- E-V2 / E-V3 FaceNet and YOLO: SKIPPED without extras
- E-V4 `cam04` + FaceNet still 0 person events: MEASURED
- Log: `05_Output/experiments/EXPERIMENT_LOG.md`

Eighty thousand cameras (DESIGN TARGET)

- About 45,000 public-domain cameras run analytics at 1 fps, 720p crop $\approx$ 80 KB
- Naive centre: $45{,}000 \times 80\,\mathrm{KB/s} \approx 3.6\,\mathrm{GB/s}$
- Five regions (Ahmedabad, Surat, Rajkot, Vadodara, Bhuj): $\approx$ 720 MB/s each
- SOC wall pulls 16–64 HLS tiles. Everything else is metadata + crops
- These numbers are design targets, not measured laptop throughput

MEASURED on this laptop (04 Sep 2026)

- Four open file captures succeed. The fifth is rejected
- Mean JPEG from eight own-feed stills: 41 KB
- `/api/health` p99: 9.4 ms over 50 calls
- GPU count: 0. Regional GPUs stay DESIGN TARGET
- Statewide 80 KB crop remains DESIGN TARGET (ratio 1.9$\times$)

Cost of the intelligence plane (DESIGN TARGET)

Item (statewide, per year)	INR
5 regional GPU nodes (3-yr amortised)	1.8–2.4 Cr
Object storage for crops + selected video	0.8–1.2 Cr
SWAN / leased bandwidth top-up	0.6–1.0 Cr
Integration + SOC ops (12 engineers)	1.5 Cr
Total intelligence plane	$\sim$5–6 Cr

- This is not the bill for replacing 26 VMS estates

Capture internals

- `StreamSession` is the only OpenCV capture holder
- TCP flag is set before `import cv2`
- Event time is PTS. Wall clock is only for reconnect backoff
- `MAX_OPEN_CAPTURES` default 4. Fifth open raises
- Scene cut: PTS rewind or gap $> 5\,\mathrm{s}$. Object and face tracks reset
- File protocol is `own_feed.mp4` on `CAM-OWN-001`

RBAC matrix

User	Role	Sees	Write
`judge`	soc_operator	all	yes
`admin`	superadmin	all	yes
`home.viewer`	dept_viewer	Home only	403
`auditor`	auditor	all	403

- Mall camera is hidden from Home viewer
- Enroll and confirm-face are write routes

Path jail and HMAC tiles

- Crops resolve inside media roots. `..` is rejected
- Stream token: HMAC over camera, actor, expiry. TTL 60 s
- Full digest, `compare_digest`. Not a sliced hash
- Browser never receives `rtsp://`
- `hls.min.js` is vendored. No jsDelivr
- Sentinel cookie stays on the server

What we took, what we refused

- Take: FaceNet, YOLO with person, ByteTrack, enroll form, original crops, predict
- Refuse: FaceAnalyzer on every camera
- Refuse: 80k footer without DESIGN TARGET
- Refuse: catalogue `/api/ingest` (live 404)
- Refuse: torch in default requirements
- Refuse: branded reconstructor names, Wikipedia tests, changing origin

Security in the PoC

- Roles: `superadmin`, `soc_operator`, `dept_viewer`, `auditor`
- `home.viewer` sees Home cameras only; private-permitted mall is hidden
- HLS/file tiles use a 60 s HMAC token. Raw RTSP never reaches the browser
- Sentinel cookie stays on the server. The browser never sees it
- Private-permitted onboard requires `consent=true`
- Audit: onboard, watchlist, ack, report, plate confirm, face confirm

Stack (open source, as required)

Piece	PoC	Later swap
API	FastAPI + Uvicorn	same contract
UI	Leaflet + vanilla JS	React if needed
Store	SQLite	PostgreSQL + PostGIS
Capture	OpenCV + FFmpeg, RTSP TCP	DeepStream
ANPR	morphology + Tesseract	YOLO + PaddleOCR
Alerts	in-process WebSocket	Redis / Kafka

What is done, what is still human

- Done: platform, HLD, this deck, live catalogue sync, RTSP frame, confirm CSV
- Done: objects, intrusion, lawful FRS with gov refuse, optional engine pack
- Done: GitHub `github.com/amitduabits/PRAHARI`
- Human: own-feed YouTube $\leq$ 3 min. Gov-feed YouTube $\leq$ 3 min, no FRS
- Human: Drive Anyone+Viewer for `gov_feed_plates.csv`
- Human: Tesseract on PATH if OCR should appear on camera
- Stop product edits after the portal receipt on 07 Sep

Demo run (3 minutes)

- Open `http://127.0.0.1:8080`. Login `judge`
- Operations map: seeded pins, Dahod offline on purpose
- Cameras table: Open tile on `cam04` or `cam01`
- Vehicle Track: reconstruct `GJ01AB1234`, download CSV
- Alerts: CRITICAL queue, Ack writes audit
- Onboard: Analyse this still, or Operator confirm if OCR misses
- Gaps: short retention on Food and Civil Supplies